BankFlow

A Modern Mobile Banking Application

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Mobile Application Development Course

1. Introduction

BankFlow is a comprehensive mobile banking application developed using Flutter and Supabase. The application demonstrates real-world banking functionality while maintaining a clean, intuitive user interface. This report presents the concept, technical implementation, and database architecture of the application.

The primary objective was to create a cross-platform financial management solution that enables users to manage their accounts, perform money transfers, track transactions, and analyze their spending patterns. Flutter was chosen for its ability to compile to native code for both iOS and Android from a single codebase, while Supabase provides a robust backend-as-a-service solution with built-in authentication and real-time capabilities.

2. Application Features

The application incorporates several core banking features designed to provide a complete financial management experience:

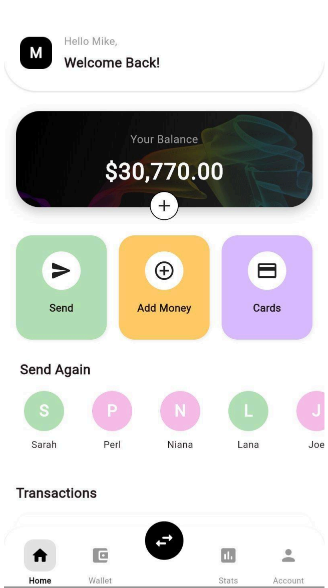
Authentication & Security: Users can register with email and password, with automatic session management ensuring persistent login states. All data access is protected through Supabase Row Level Security policies.

Money Transfers: The app enables instant peer-to-peer transfers by email address, with real-time balance updates and transaction confirmations. Users can quickly resend to frequent contacts through the "Send Again" feature.

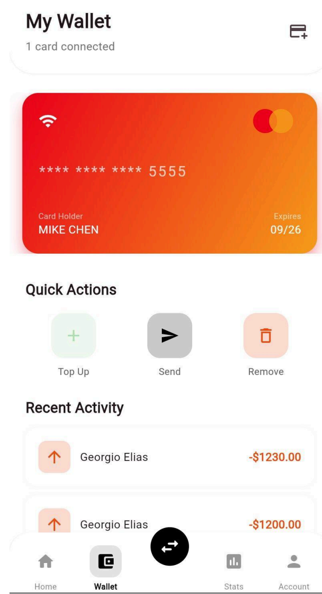
Card Management: Users can link Visa and Mastercard cards, view them in an interactive carousel, set primary cards, and manage daily spending limits.

Financial Analytics: The Statistics module provides visual charts showing income versus expenses, net flow calculations, and time-based filtering for weekly, monthly, or yearly views.

Application Screenshots



Home Dashboard



Card Management

3. Technology Stack

The application leverages modern technologies to ensure performance, scalability, and maintainability. On the frontend, **Flutter** serves as the cross-platform framework, using **Dart** as the programming language. State management is handled through the **Provider** pattern, while data visualization utilizes the **FL Chart** library for rendering interactive graphs.

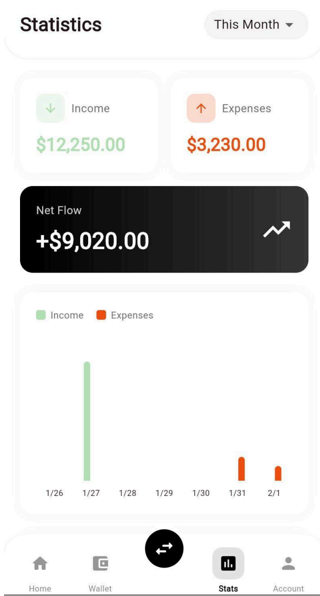
The backend infrastructure is built on **Supabase**, an open-source Firebase alternative that provides authentication, real-time databases, and storage. The database engine is **PostgreSQL**, offering robust relational data management with advanced security through Row Level Security (RLS) policies.

4. Database Architecture

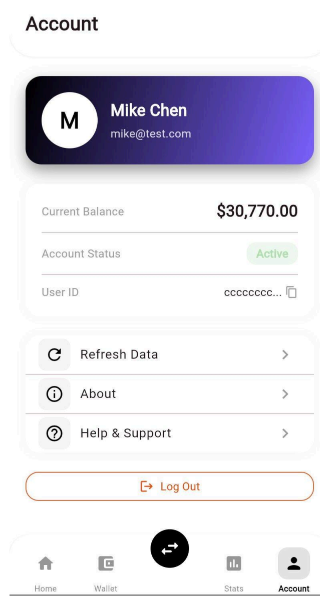
The application utilizes a PostgreSQL database hosted on Supabase with four primary tables:

Table	Purpose
`users`	User profiles, account balances, contact information, and verification status
`transactions`	Money transfers, deposits, withdrawals, and payments with status tracking
`cards`	Linked payment cards with type, holder details, expiry, and daily limits
`notifications`	User alerts and transaction notifications with read status

All tables implement Row Level Security (RLS) policies, ensuring users can only access their own data. The transactions table uses CHECK constraints to validate transaction types (send, receive, deposit, withdraw, payment) and statuses (pending, completed, failed, cancelled).



Financial Statistics



Account Management

5. Conclusion

BankFlow successfully demonstrates the implementation of a feature-rich mobile banking application using modern development practices. The combination of Flutter for cross-platform development and Supabase for backend services provides a scalable, secure, and maintainable architecture. Future enhancements could include biometric authentication, push notifications, QR code payments, and multi-currency support.

Repository: github.com/georgioelias/BankFlow